REPORT 1 -
NOSE CONE ANALYSIS

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Weisshorn – Nose Cone Design Report

EPFL ROCKET TEAM

Contents

Summary 1

Nomenclature 2

1 Inputs, Requirements and Assumptions 3
- 1.1 What is fixed . . . 3
- 1.2 What is assumed . . . 4
- 1.3 What could not be found . . . 4

2 Design Options 5
- 2.1 Supersonic flow over a nose and wave drag . . . 5
- 2.2 Candidate profiles . . . 5
- 2.3 Comparison criteria . . . 6

3 Method 6
- 3.1 Nose wave drag: axisymmetric Euler solver . . . 8
- 3.2 Transonic range . . . 9
- 3.3 Flight model . . . 9
- 3.4 Heating . . . 10
- 3.5 Stability, mass and payload . . . 10

4 Results 10
- 4.1 Nose wave drag . . . 10
- 4.2 Flight performance . . . 12
- 4.3 Trade study . . . 12
- 4.4 Robustness of the ranking . . . 14
- 4.5 Tip bluntness . . . 14
- 4.6 Heating . . . 15

5 Design Catalogue 16
- 5.1 Literature cross-check . . . 20

6 Selection 21
- 6.1 Profile . . . 21
- 6.2 Length . . . 21
- 6.3 Tip . . . 22

7 Recommendations 22
- 7.1 Numbered recommendations . . . 22
- 7.2 When to choose another design . . . 23

8 Design Definition 23
- 8.1 Manufacturing plan . . . 24
- 8.2 Mass budget . . . 25
- 8.3 CAD deliverables . . . 25

9 Open Points and Next Steps 25

A Scripts and data 26

Summary

Proposed Weisshorn nose cone	
Profile	Von Kármán (Haack $C = 0$), tangent to the body
Length	$L = 810$ mm ($L/D = 4.5$) to the theoretical apex; 803.3 mm physical
Diameter	180 mm
Tip	spherical, $r_n = 4.5$ mm ($r_n/R = 0.05$), machined Al 6082-T6 insert
Shell	CFRP $[0/45/0]_{SO}$, 1.2 mm (GFRP if the nose is the RF window), ≈354 g with tip
Fallback	same profile at $L/D = 4$ (720 mm) if the 62 mm cannot be found in the 5.4 m budget

The Von Kármán profile is the member of the Haack family of noses that, according to slender-body theory, gives the lowest supersonic pressure drag for a given length and base diameter (Section 2.2). L/D is the *fineness ratio*, the nose length divided by the body diameter. Symbols and abbreviations are listed in the Nomenclature.

Why:

- **Drag.** Above Mach 1 a shock wave forms ahead of the nose, and the overpressure it creates on the nose surface produces a pressure drag called *wave drag*. Wave drag is the only part of the vehicle drag that depends on the nose shape (Section 2.1). Seven profiles (cone, tangent ogive, parabola, power $\frac{1}{2}$ and $\frac{3}{4}$, LV-Haack, Von Kármán; Section 5) were compared with an axisymmetric Euler solver, i.e. a CFD code for inviscid compressible flow (validated to $< 0.5\,\%$ against Taylor–Maccoll, the exact solution for a cone, and grid-converged to 0.5 %). The solver was coupled to a trajectory model calibrated on the Launch Vehicle Overview (LVO). At $L/D = 4.5$, Von Kármán has the lowest wave drag at every computed Mach number. It flies 18 m to 22 m higher than the power noses, 69 m to 79 m higher than ogive, parabola and LV-Haack, and 132 m higher than the cone. It stays first in all seven robustness scenarios. At $-143\,\mathrm{m/kg}$ (apogee lost per kilogram of dry mass), the margin over the ogive is worth about 0.55 kg of dry mass.
- **Length.** $L/D = 4.5$ adds 60 m over the LVS length (the 720 mm nose of the Launch Vehicle Sizing) for 62 mm of nose bay, with diminishing returns beyond that: wave drag falls as $1/(L/D)^2$, while the friction and the mass of the nose grow linearly with its length.
- **Heating.** T_0, the stagnation temperature (the temperature air reaches when it is brought to rest against the tip, and the upper bound for any surface temperature), peaks at 119 °C (nominal) and 156 °C (hot-day envelope, $M = 1.56$). A printed PET-CF tip would exceed its heat-deflection temperature (the temperature at which the polymer starts to deform under load); the Al tip stays below 73 °C. The shell inner face reaches 104 °C in the envelope, close to the limit of the DP490 epoxy that bonds the coupler to the shell, so the thermal requirement must be raised from Firehorn's 390 K/5 s.
- **Integration and stability.** The tangent shoulder (no change of slope where the nose meets the body tube) suits the bonded coupler; 254 mm of the 3U payload fits inside the curved part. The static margin, the distance from the centre of gravity to the centre of pressure in body diameters (calibres, cal), changes by -0.04 cal, which is negligible.
- **Deliverables.** Profile coordinates, watertight STL solids of all seven designs and of the final

nose, a STEP solid of the final nose, the full Euler data set, nine numbered recommendations with owners (Section 7) and the alternatives to use if the priorities change.

Not found in the ERT documentation (details in Section 1.3): Alex Marseille's nose cone report, any Weisshorn nosecone requirement, the Weisshorn fin geometry and mass breakdown, and any supersonic flight data.

Nomenclature

Symbol	Meaning	Unit
Geometry		
D, R	body (base) diameter 180 mm, radius $R = D/2$	m
L, L/D	nose profile length (theoretical apex to shoulder); fineness ratio	m, –
x, y; ξ, η	axial coordinate from the apex, local radius $y(x)$; $\xi = x/L$, $\eta = y/R$	m; –
r_n	radius of the spherical tip (bluntness r_n/R)	m
A_b	base area πR^2, reference area of all nose coefficients	m^2
V	internal volume of the nose (in Eq. (3): cell volume per radian)	m^3
θ_{sh}	angle between the profile and the body tube at the shoulder	°
n, K', C, θ	power-series exponent; parabolic-series parameter; Haack-series parameter and angle variable	–
Flow		
M, M_∞, $M_{\max}$	Mach number V/a; free-stream Mach number; maximum of the flight	–
V_∞, a	flight speed; speed of sound $\sqrt{\gamma R_g T}$	m/s
γ, R_g	ratio of specific heats (1.4); gas constant of air (287 J/(kg K))	–, J/(kg K)
μ	Mach angle, $\sin\mu = 1/M$	°
p, ρ, T	static pressure, density (in Table 2: ogive arc radius), temperature; subscript ∞: free stream	Pa, kg/m^3, K
u, v, E	axial and radial velocity; total energy per unit volume	m/s, J/m^3
q_∞	dynamic pressure $\frac{1}{2}\rho_\infty V_\infty^2$	Pa
C_p	pressure coefficient $(p - p_\infty)/q_\infty$	–
Aerodynamic coefficients and flight		
$C_{D,\mathrm{w}}$, C_D	nose wave-drag coefficient (reference A_b); drag coefficient of the vehicle	–
$C_{D,f}$, $C_{D,\mathrm{fins}}$, $C_{D,\mathrm{base}}$, $C_{D,e}$	skin-friction, fin pressure, base and excrescence drag coefficients	–
k	calibration factor on the drag of everything except the nose	–
$C_{N\alpha}$	normal-force slope $\mathrm{d}C_N/\mathrm{d}\alpha$ (α: angle of attack)	rad^{-1}
x_{cp}, x_{cg}	position of the centre of pressure and of the centre of gravity	m
F_{th}, m, g_0	thrust; vehicle mass; standard gravity	N, kg, m/s^2
I_t, I_{sp}	total impulse; specific impulse	N s, s
z, Δz	altitude; apogee change relative to the baseline nose	m
Heat transfer		
T_0, T_r, T_w	stagnation, recovery and wall temperature; r: recovery factor	K
h, h_s	convective heat-transfer coefficient; its value at the stagnation point	$W/(m^2\,K)$
c_p; Pr, Re_x, St	specific heat; Prandtl, Reynolds (on the running length x) and Stanton numbers	J/(kg K); –

Abbreviations. 3U: CubeSat payload unit, $100 \times 100 \times 340.5$ mm; cal: calibre (length in body diameters); CFD: computational fluid dynamics; CFL: Courant–Friedrichs–Lewy number; CFRP, GFRP: carbon-, glass-fibre-reinforced polymer; CG, CP: centre of gravity, of pressure; DJF, DDF: Design Justification, Definition File; DP490: 3M Scotch-Weld epoxy of the coupler bonds; EuRoC: European Rocketry Challenge; FD, ST, SE, AV, PL: ERT Flight Dynamics, Structure, Systems Engineering, Avionics, Payload teams; GCI: grid convergence index; HLLC: Harten–Lax–van Leer–Contact Riemann solver; ISA: International Standard Atmosphere; LV, LVO, LVS: launch vehicle, Launch Vehicle Overview, Launch Vehicle Sizing; MUSCL: Monotonic Upstream-centred Scheme for Conservation Laws; PDR, CDR: Preliminary, Critical Design Review; PET-CF: carbon-fibre-filled PET filament; RANS: Reynolds-averaged Navier–Stokes; RF: radio frequency; SSP-RK2: two-stage strong-stability-preserving Runge–Kutta.

1 Inputs, Requirements and Assumptions

1.1 What is fixed

Table 1 lists every input taken from the ERT documentation. The vehicle is **Weisshorn**, the 2027 competition launch vehicle flying in the EuRoC L9 category (target apogee 9 km). The sources are the Launch Vehicle Overview (LVO) and Launch Vehicle Sizing (LVS) of systems engineering, the requirement database (REQ), and the Design Justification and Definition Files (DJF, DDF) of earlier vehicles.

Table 1: Inputs from the ERT wiki, the Weisshorn requirements database and the Firehorn documentation.

Quantity	Value	Source
External diameter	180 mm	task; 2027_C_SE_REQ_03
Mission	EuRoC 2027, L9, 9 km apogee	Weisshorn wiki page; 2027_C_SE_LVO
LV length	$<$ 5.4 m (LVO: $\sim$ 5.4 m)	2027_C_SE_REQ_04; 2027_C_SE_LVO
Dry / wet mass	$\sim$ 45 kg / $\sim$ 75 kg ($<$ 50 kg dry)	2027_C_SE_LVO; 2027_C_SE_REQ_06
Engine	EthaLOx, 4.5 kN, 9.1 s, $I_t = 40\,960$ N s, $I_{sp} = 200$ s	2027_C_SE_LVO; 2027_C_SE_LVS_DJF
Maximum speed	Mach 1.4	2027_C_SE_LVO
Stability	1.5–6 cal (EuRoC); FD target 2–4 cal	2027_C_SE_REQ_09; 2027_C_SE_FD_REQ_07
Launch rail	$84 \pm 1^\circ$ elevation	2027_C_SE_FD_REQ_05
Payload	3U CubeSat in the nosecone bay, nose separates at apogee	2027_C_SE_ST_FUA; Structure wiki page
Nose length (sizing)	scaled from Firehorn: $960 \times 180/240 = 720$ mm	2027_C_SE_LVS_DJF (“scaled down in the same way for mass and length”)
RF windows	360°, $\geq 2D$ long	2027_C_SE_AV_INTERFACE_ST_REQ_07
Firehorn manufacturing process	CFRP prepreg on PET-CF male mould, $[0/45/0]_{SO}$, 1.2 mm; removable PET-CF tip; bonded Al male coupler; as-built 500 g	2024_C_ST_NOSECONE_DJF/DDF; 2025_C_ST_NOSECONE_PURPOSE_IR
Firehorn thermal requirement	390 K for 5 s	2024_C_SE_ST_NOSECONE_REQ_09 (Firehorn)

With $M_{\max} \approx 1.4$, Weisshorn crosses the *transonic* range (roughly $0.8 < M < 1.2$, where subsonic and supersonic regions coexist around the body) and flies a few seconds in the low supersonic range, so the nose is designed for this whole range rather than for one Mach number. The 3U CubeSat ($100 \times 100 \times 340.5$ mm) sits in the nose bay, so the profile decides how much of it fits inside the curved part. Carbon fibre blocks radio signals, so if the nose has to be the RF window its shell must be glass fibre.

1.2 What is assumed

Each assumption is chosen so that the comparison between nose shapes is fair; none of them changes the ranking.

- **Trajectory.** Point mass (thrust, drag and gravity; the rotation of the stable vehicle is ignored), rail 12 m at $84°$, ISA, no wind, constant 4.5 kN for 9.1 s. Propellant from $I_t/(I_{sp}g_0) = 20.9$ kg, i.e. burn-out mass 54.1 kg. The drag of everything except the nose, the least certain part of the model at this stage, is scaled by a single factor k so that the baseline nose reaches the LVO apogee of 9.0 km (Section 3.3). This gives $M_{\max} = 1.47$, consistent with the LVO's Mach 1.4. All noses share the same k, and Section 4.4 shows that the ranking does not change when the rest of the drag is varied.
- **Thermal envelope.** The LVO masses ($75 \rightarrow 45$ kg) on a 30 °C day give $M_{\max} \approx 1.56$, the hottest credible flight. Heating is checked on this case.
- **Fins** (not yet defined for Weisshorn): four trapezoidal fins, root 300 mm, tip 120 mm, span 160 mm, sweep 180 mm, 5 mm hexagonal section. They only enter the common drag and the CP shift.
- **Payload envelope.** A 3U unit needs an outer radius $r \geq 77$ mm: 70.7 mm half-diagonal, 3 mm clearance, 2 mm guide, 1.2 mm shell.
- **Shell.** 1.2 mm CFRP at the Firehorn as-built areal density (0.98 kg/m^2), plus a 60 g machined aluminium tip.
- **Transonic nose wave drag** ($0.8 < M < 1.1$): smooth rise from zero at $M = 0.8$ to the computed $M = 1.1$ value at $M = 1.0$ (Section 3.2).

1.3 What could not be found

- **Alex Marseille's nose cone report** is not on the ERT Drive or the wiki. His documented work is the CFD of the 54 mm supersonic semester rocket; as CFD advisor he set the semester goal "WS (Cd, Nosecone)" (2027_C_FD_TM_3), which is this study. The prior work used instead is the 2021 supersonic nosecone semester project, the Wildhorn optimisation note (2022_FD_TN_C01) and the Firehorn nosecone DJF/DDF.
- **No Weisshorn nosecone requirement** (length, mass, thermal) exists yet; the Weisshorn nosecone pages are empty templates.
- **Weisshorn fins, CG and mass breakdown** are not defined. The static margin is therefore given as a *change* relative to the baseline nose.
- **No supersonic flight data** exists at ERT to calibrate drag or heating against.

2 Design Options

2.1 Supersonic flow over a nose and wave drag

Above $M = 1$, pressure disturbances can no longer travel upstream: a disturbance created at a point only influences the region inside a downstream cone of half-angle

$$\mu = \arcsin \frac{1}{M}, \tag{1}$$

the *Mach angle* ($\mu = 45.6°$ at $M = 1.4$). The air ahead of the nose receives no advance signal and is deflected abruptly through a *shock wave*, across which pressure, density and temperature jump. Where the surface turns back towards the axial direction, the flow *expands* and the pressure falls, possibly below ambient. The pressure on a supersonic nose is therefore above ambient near the tip and decreases along it. Since the nose faces forward, this gives a net axial force, the *wave drag*, which exists even in an inviscid fluid. It is expressed as $C_{D,\mathrm{w}} = D_\mathrm{w}/(q_\infty A_b)$ with the base area $A_b = \pi R^2$ (Equation (4)). Wave drag appears once the local flow becomes sonic somewhere on the nose, around $M \approx 0.8$ for slender bodies, peaks near $M = 1$ and decreases slowly in the supersonic range. It is the only drag component controlled by the nose shape.

A sharp, slender tip deflects the flow by a small angle and carries a weak oblique shock attached to the tip. A blunt tip, or a half-angle too large for the Mach number, makes the shock detach into a curved *bow shock* whose central, normal part brings the flow almost to the stagnation pressure. This is why supersonic noses are pointed and why any rounding of the tip has to stay small (Section 4.5).

For a slender nose, linearised (slender-body) theory makes the wave drag depend only on the axial distribution of cross-sectional area $S(x) = \pi y(x)^2$, through $S''(x)$. Within a profile family $S'' \propto R^2/L^2$, so

$$C_{D,\mathrm{w}} \propto \left(\frac{R}{L}\right)^2 \propto \frac{1}{(L/D)^2} : \tag{2}$$

doubling the length divides the wave drag by four. The area distribution of minimum wave drag is also known: for a given length and base area it is the Von Kármán profile, with $C_{D,\mathrm{w}} = 1/(L/D)^2$; for a given length and volume it is the LV-Haack profile. The theory gives trends rather than exact values and is least accurate near $M = 1$; the Euler computations of Section 3 provide the values used in this study.

2.2 Candidate profiles

The nose is defined by three choices: the **profile family**, the **fineness** L/D and the **tip bluntness** r_n/R. Seven profiles are compared (Table 2, Figure 1). With $\xi = x/L$ and $\eta = y/R$:

- **Conical**: straight generator of half-angle $\arctan(R/L)$ ($7.1°$ at $L/D = 4$), meeting the body with a kink of the same angle.
- **Tangent ogive**: circular arc of radius $\rho = (R^2 + L^2)/(2R)$, tangent to the body at the shoulder; the traditional subsonic rocket nose.
- **Parabolic series**: the case $K' = 1$ is the parabola tangent to the body at the shoulder.

- **Power series** $\eta = \xi^n$: the smaller n, the fuller the front. For $n = \frac{1}{2}$ the apex has a radius of curvature $R^2/(2L)$ (5.6 mm at $L/D = 4$), so this nose is blunt by construction. Power noses meet the body with a kink $\theta_{\text{sh}} = \arctan(nR/L)$.
- **Haack series**: the minimum-wave-drag area distributions of slender-body theory, with $C = 0$ (**Von Kármán**, given length and base area) and $C = \frac{1}{3}$ (**LV-Haack**, given length and volume). Both are tangent at the shoulder; near the apex the Von Kármán profile grows as $\xi^{3/4}$ (vertical tangent, zero tip radius).

The profiles differ in three features (Figure 1): how the nose meets the body (tangent, or with a kink), the shape of the tip (finite-angle point, pointed with a vertical tangent, or blunt) and how the area is distributed along the length, from the thin cone to the full LV-Haack nose. The elliptical nose (strong bow shock above Mach 1) and the secant ogive (no advantage over the tangent ogive and LV-Haack, kinked shoulder) were discarded before computing.

Table 2: Candidate profiles (ogive in dimensional form, ρ: arc radius). x_{cp}: Barrowman nose centre of pressure; θ_{sh}: angle to the body at the shoulder (0 = tangent).

Profile	$\eta(\xi)$	ERT vehicles	x_{cp}/L	θ_{sh} at $L/D = 4$
Conical	ξ	–	0.667	$7.1°$
Tangent ogive	$\sqrt{\rho^2 - (L-x)^2} + R - \rho$, $\rho = \frac{R^2+L^2}{2R}$	Bella Lui II, Nordend	0.464	$0°$
Parabolic ($K' = 1$)	$2\xi - \xi^2$	–	0.467	$0°$
Power series $n = \frac{1}{2}$	$\xi^{1/2}$	Wildhorn, Firehorn	0.500	$3.6°$
Power series $n = \frac{3}{4}$	$\xi^{3/4}$	–	0.600	$5.4°$
LV-Haack ($C = \frac{1}{3}$)	$\pi^{-1/2}\sqrt{\theta - \frac{1}{2}\sin 2\theta + C\sin^3\theta}$, $\theta = \arccos(1-2\xi)$	–	0.438	$0°$
Von Kármán ($C = 0$)	as LV-Haack with $C = 0$	–	0.500	$0°$

2.3 Comparison criteria

- **Wave drag.** At Weisshorn's Mach numbers it depends mostly on L/D and on how much area the profile adds near the tip. Slender-body theory makes Von Kármán the favourite; the Euler computations check whether this holds at Weisshorn's finite thickness and Mach numbers.
- **Volume and payload.** Fuller profiles reach $r = 77$ mm further forward, so more of the 3U sits inside the curved part. The rest needs a cylindrical extension of the nose bay, which adds vehicle length within the 5.4 m cap.
- **Stability.** Slender-body theory gives every nose that closes on A_b the same normal-force slope $C_{N\alpha} = 2$ per radian; the shape only sets where the force acts, $x_{cp} = L - V/A_b$ from the apex ($2L/3$ for a cone). A fuller nose moves the vehicle CP forward and lowers the static margin $(x_{cp} - x_{cg})/D$.
- **Manufacture.** A tangent shoulder ($\theta_{\text{sh}} = 0$) avoids a kink at the coupler bond line, a fillet in the mould and a local expansion in the flow. A sharp tip cannot be laid up in prepreg, so every option uses a separate machined tip.

3 Method

The chain is: Euler wave drag of each nose → vehicle drag build-up → ascent trajectory → apogee, heating, stability. The wave drag is the only shape-dependent quantity; the flight model

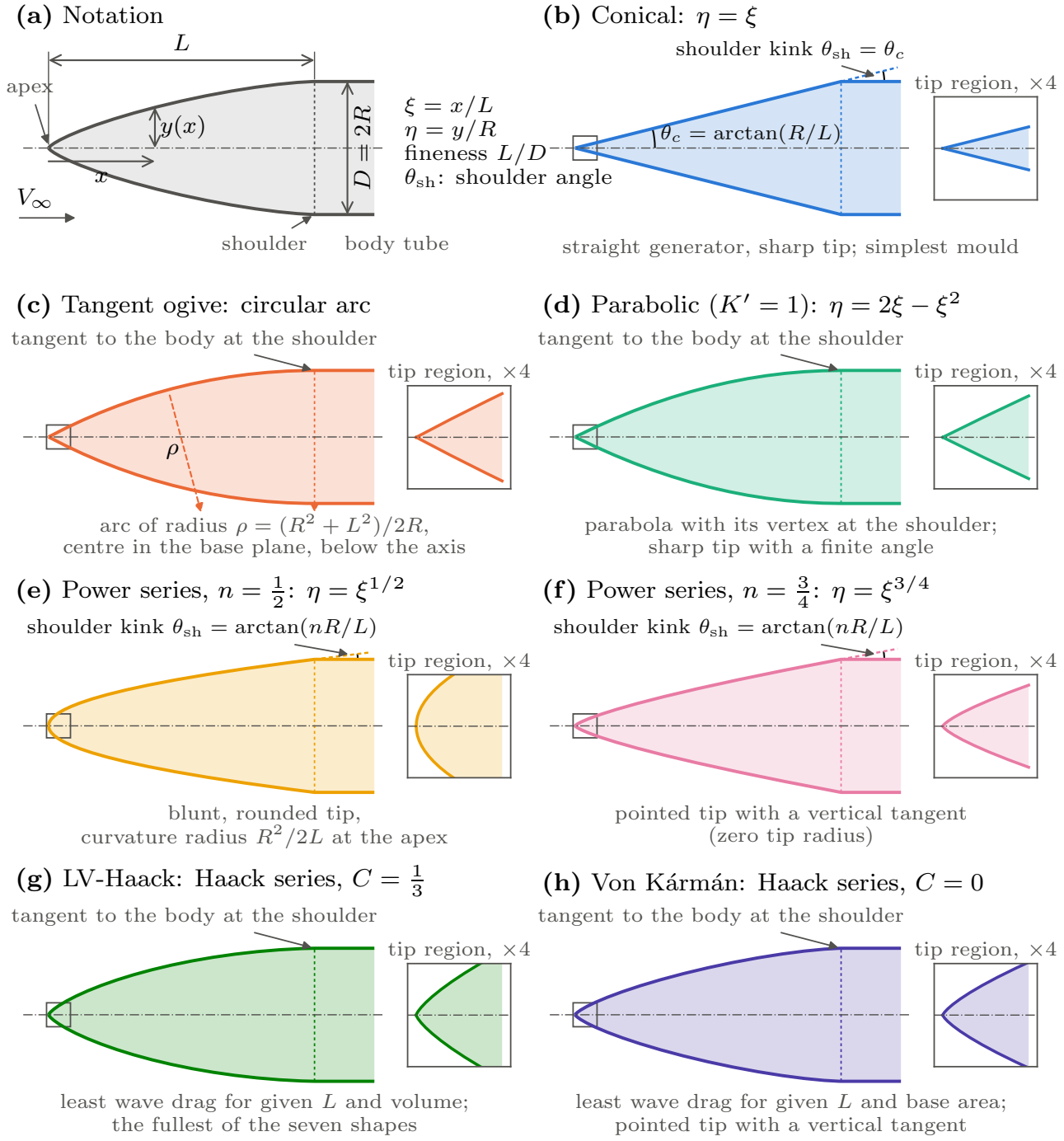


Figure 1: Schematics of the candidate profiles, drawn at $L/D = 2$ (half the design fineness) so that their differences are visible; Figure 10 shows them at real proportions. (a) Notation. (b)–(h) Construction of each profile: the upper label describes the shoulder (tangent to the body, or a kink of angle θ_{sh}), the lower label the generator and the tip, and the inset shows the boxed tip region magnified four times.

converts it into apogee and also accounts for the length and mass of each nose. The implementation is described in Appendix A.

3.1 Nose wave drag: axisymmetric Euler solver

Above Mach 1 the nose pressure drag is set by the inviscid outer flow (shock and expansion), so the axisymmetric Euler equations (the Navier–Stokes equations without viscous terms) are solved for each nose followed by a $5R$ cylinder. The boundary layer is thin at these Reynolds numbers (about 2×10^7 on the nose length at $M_{\max}$); its main effect, skin friction, is added in the flight model (Section 3.3). In finite-volume form, per radian of azimuth,

$$\frac{\partial (\boldsymbol{U} V)}{\partial t} + \sum_{\text{faces}} \boldsymbol{F} \cdot \boldsymbol{n}\, \ell\, r_f = \begin{pmatrix} 0 & 0 & p\,A & 0 \end{pmatrix}^{\top}, \qquad V = A\, r_c, \tag{3}$$

with $\boldsymbol{U} = (\rho, \rho u, \rho v, E)$ the conserved variables, $\boldsymbol{F}$ their flux through a face of outward normal $\boldsymbol{n}$, length ℓ and mid-radius r_f, and $V = A r_c$ the volume per radian of a ring-shaped cell of area A and centroid radius r_c. The source $p\,A$ is the net radial pressure force on the two non-parallel azimuthal faces of the cell; writing the volume as $A r_c$ preserves a uniform flow exactly.

The scheme is a cell-centred finite-volume method with HLLC approximate Riemann fluxes (shocks resolved within two to three cells without oscillations), MUSCL reconstruction of the primitive variables with the minmod limiter (second order away from shocks), and SSP-RK2 time integration with local time stepping at CFL 0.85, since only the steady state matters. The body-fitted grid (160×80 cells) starts on the stagnation streamline upstream of the tip and clusters points at the tip and at the shoulder. The outer boundary lies outside the bow shock with characteristic (non-reflecting) far-field conditions; the wall is a slip wall and the axis a symmetry line. The drag coefficient is the integral of the wall overpressure over the nose,

$$C_{D,\mathrm{w}} = \frac{1}{q_\infty \pi R^2} \int_{\text{nose}} (p - p_\infty)\,(-n_x)\, 2\pi r \,\mathrm{d}s, \tag{4}$$

where n_x is the axial component of the outward wall normal (negative on a forward-facing surface). A run is converged when $C_{D,\mathrm{w}}$ changes by less than 2×10^{-5} (relative) over 250 iterations. The Mach numbers are computed from high to low (1.6, 1.4, 1.2, 1.1; also 1.3 for the cone and ogive), each started from the previous solution. The $M = 1.1$ runs, with a large subsonic region behind the bow shock, were stopped at 6000 iterations; they only set the end point of the transonic model and carry an uncertainty of 1–2 %. The trade study interpolates $C_{D,\mathrm{w}}(L/D)^2$, nearly constant by Equation (2), between $L/D = 3$, 4 and 5. All seven shapes, and the blunted Von Kármán nose, were also computed directly at $L/D = 4.5$ for $M = 1.6$, 1.4 and 1.2. The interpolation agrees with these direct runs within 0.3 %, but scaling the parabolic and power-$\frac{3}{4}$ noses from $L/D = 4$ alone underestimated their drag by 2–4 %, so the direct values are used at $L/D = 4.5$ throughout.

Validation. For a sharp cone the flow between shock and body is conical, and the Euler equations reduce to the Taylor–Maccoll ordinary differential equation, integrated here to high accuracy. The surface pressure is uniform, so $C_{D,\mathrm{w}} = C_p$. Figure 2a compares the two for four cone angles ($7.1°$ to $20°$, $M = 1.1$–2): the largest error is 0.46 % (the $20°$ cone at $M = 1.2$ has a detached shock and no conical solution). The grid convergence study (Figure 2b) uses the Von Kármán nose at $L/D = 4$ on three grids (80×40, 160×80 production, 320×160). At $M = 1.4$ the observed order is $p = 2.0$, the design order of the scheme, and the grid convergence

index (GCI, Roache, safety factor 1.25) of the production grid is 0.5 %; at $M = 1.2$ the two-grid estimate ($p = 2$, safety factor 3) gives 1.1 %. The production grid is therefore accurate to about 0.5–1 %, well below the 3–46 % by which the other shapes exceed the Von Kármán wave drag at $L/D = 4.5$.

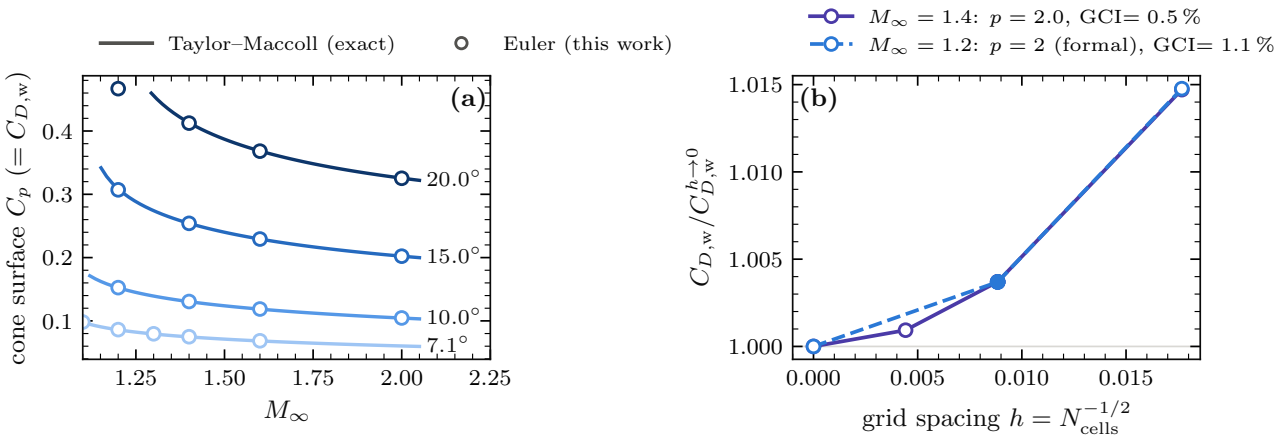


Figure 2: (a) Cone surface pressure from the Euler solver (markers) against the exact Taylor–Maccoll solution (lines). (b) Grid convergence of the nose wave drag, normalised by the Richardson-extrapolated value (open: computed grids; filled: production grid).

3.2 Transonic range

Steady Euler solutions converge too slowly below $M = 1.1$, and near $M = 1$ the shocks interact with the boundary layer, so viscous (RANS) computations are needed there anyway. The nose wave drag is therefore modelled, for all shapes, as a smooth cubic rise from zero at $M = 0.8$ (first sonic flow on a slender nose) to the computed $M = 1.1$ value at $M = 1.0$, flat up to $M = 1.1$. The Fluent runs of Section 9 replace it; until then its onset and amplitude are varied in the robustness study (Section 4.4).

3.3 Flight model

A point-mass trajectory is integrated from the 12 m rail ($84°$) to apogee. Along the flight path, with V_∞ the speed and γ_f the flight-path angle,

$$m\frac{\mathrm{d}V_\infty}{\mathrm{d}t} = F_{\mathrm{th}} - \tfrac{1}{2}\rho V_\infty^2 A_b C_D - mg\sin\gamma_f, \qquad V_\infty\frac{\mathrm{d}\gamma_f}{\mathrm{d}t} = -g\cos\gamma_f, \tag{5}$$

with the mass m decreasing linearly during the burn and ρ from the ISA. The drag coefficient (reference area $\pi D^2/4$) is built up from components,

$$C_D = k\,(C_{D,f} + C_{D,\mathrm{fins}} + C_{D,\mathrm{base}} + C_{D,e}) + C_{D,\mathrm{w}}(M)\,: \tag{6}$$

turbulent, roughness-limited skin friction on the wetted area of nose and body (Schlichting, with a compressibility correction), fin pressure drag (blunt leading edges subsonic, Ackeret thin-airfoil theory for the hexagonal section supersonic), the OpenRocket base-drag law (nozzle exit area removed while burning) and the ERT-Sim excrescence law. Only $C_{D,\mathrm{w}}$ changes from one nose to the next; a longer nose also adds wetted area to $C_{D,f}$ and shell mass to m. The factor k is set so that the baseline nose (Von Kármán, 720 mm) reaches the LVO apogee of 9.0 km; the resulting $M_{\max} = 1.47$ is close to the LVO's Mach 1.4 without further tuning.

3.4 Heating

No wall can exceed the stagnation temperature $T_0 = T_\infty(1+\frac{\gamma-1}{2}M^2)$, the temperature air reaches when brought to rest adiabatically, which is the Firehorn design rule. On the hot day at $M = 1.56$, $T_0 = 288 \times (1+0.2\times 1.56^2) = 428$ K (155 °C). How close a surface gets to it depends on the heat it receives, $q = h\,(T_r - T_w)$, and the heat it conducts away. At the tip, the laminar stagnation-point coefficient (Fay–Riddell form without dissociation, modified-Newtonian velocity gradient) is

$$h_s = 0.763\,\mathrm{Pr}^{-0.6}\sqrt{\rho_s\mu_s\left.\frac{\mathrm{d}u_e}{\mathrm{d}x}\right|_s}\,c_p, \qquad \left.\frac{\mathrm{d}u_e}{\mathrm{d}x}\right|_s = \frac{1}{r_n}\sqrt{\frac{2(p_s-p_\infty)}{\rho_s}}, \tag{7}$$

with the stagnation-point density, viscosity and pressure ρ_s, μ_s, p_s behind the normal shock. Hence $h_s \propto r_n^{-1/2}$: a larger tip radius gives a thicker boundary layer and less heating. Along the shell, the turbulent flat-plate correlation $\mathrm{St} = 0.0296\,\mathrm{Re}_x^{*-0.2}\mathrm{Pr}^{-2/3}$ (properties at Eckert's reference temperature) drives the wall towards the recovery temperature $T_r = T_\infty(1 + r\frac{\gamma-1}{2}M^2)$, $r = \mathrm{Pr}^{1/3} \approx 0.89$, the temperature of an insulated wall. The shell is modelled by 1D transient conduction through its thickness with an adiabatic inner face (conservative: no radiation or internal convection), and a metal tip as a single lumped mass heated through its wetted surface. Both are evaluated along the nominal and the hot-day envelope trajectories.

3.5 Stability, mass and payload

Barrowman's method, the standard linear method also used by OpenRocket, gives the nose $C_{N\alpha} = 2$ at $x_{cp} = L - V/A_b$ and the assumed fins $C_{N\alpha,f} = 7.4$. The vehicle CP is the $C_{N\alpha}$-weighted mean of the components, and the static margin is $(x_{cp}-x_{cg})/D$. Positions are measured from the aft end, so that fins and body stay fixed when the nose length changes, and the CG shift from the nose-shell mass is included. The shell mass is the wetted area times the Firehorn as-built areal density (0.98 kg/m^2: 500 g for the 960 mm, Ø240 mm nose). The 3U fit is the length of curved nose over which $r \geq 77$ mm.

4 Results

4.1 Nose wave drag

Figure 3a gives the wave drag of the seven profiles at $L/D = 4$ and Figure 4 the wall pressure coefficient $C_p = (p - p_\infty)/q_\infty$ behind it. By Equation (4), regions with $C_p > 0$ add drag and regions with $C_p < 0$ reduce it.

- **Von Kármán and power $n = \frac{1}{2}$ are the two low-drag profiles**, within a few per cent of each other: Von Kármán is lower at $M \geq 1.3$, power $\frac{1}{2}$ at $M \leq 1.2$. The tangent profiles expand continuously and their wall pressure falls below ambient over the aft third, which partly cancels the overpressure near the tip. The power-$\frac{1}{2}$ nose concentrates its overpressure in a short spike at its blunt tip and stays just above ambient behind it.
- **Ogive, parabolic and LV-Haack** carry 13–22 % more wave drag: their fuller forebody compresses the flow harder near the tip (higher C_p between $x/L \approx 0.1$ and 0.4, Figure 4b).
- **The cone is the worst** at every Mach number, and its drag rises steeply towards Mach 1: its constant surface angle keeps the conical overpressure all the way to the shoulder.

- **Fineness.** $C_{D,\mathrm{w}}$ falls as $1/(L/D)^2$ for every profile, the slope of slender-body theory (Figure 3b). For Von Kármán the Euler level is about 15 % below the linear-theory value, so linear theory gives the trend but not the level.

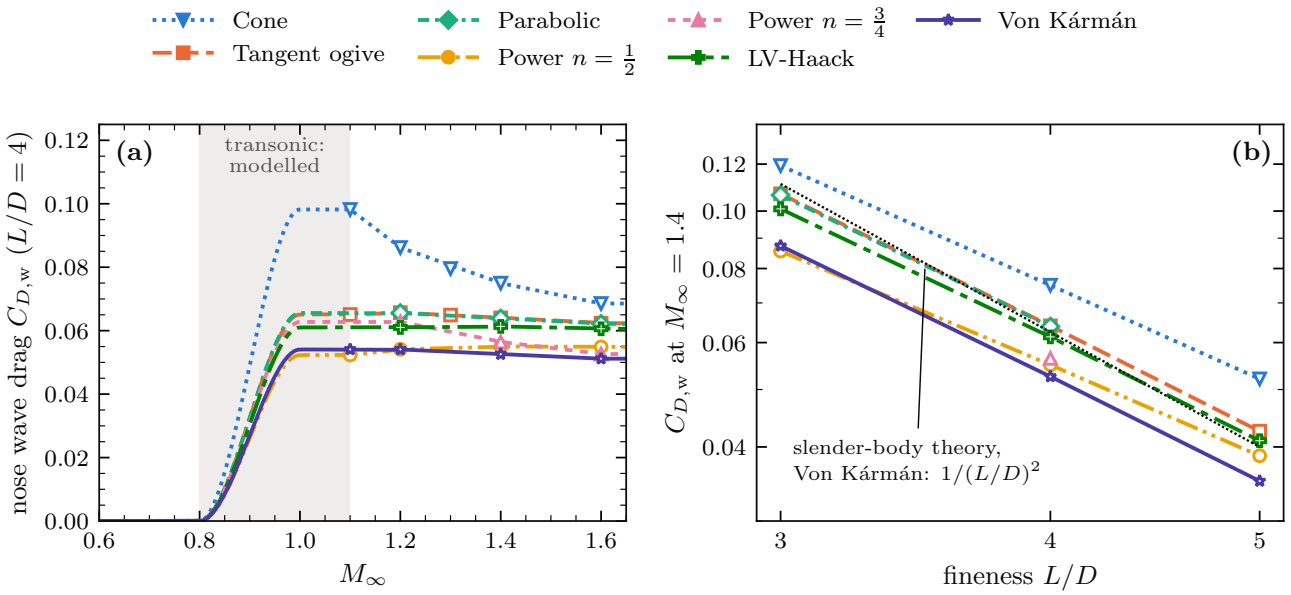


Figure 3: (a) Nose wave drag versus Mach number at $L/D = 4$ (markers: Euler; lines: model used in the trajectory; grey band: transonic model). (b) Wave drag versus fineness at $M = 1.4$, with the slender-body result for the Von Kármán nose.

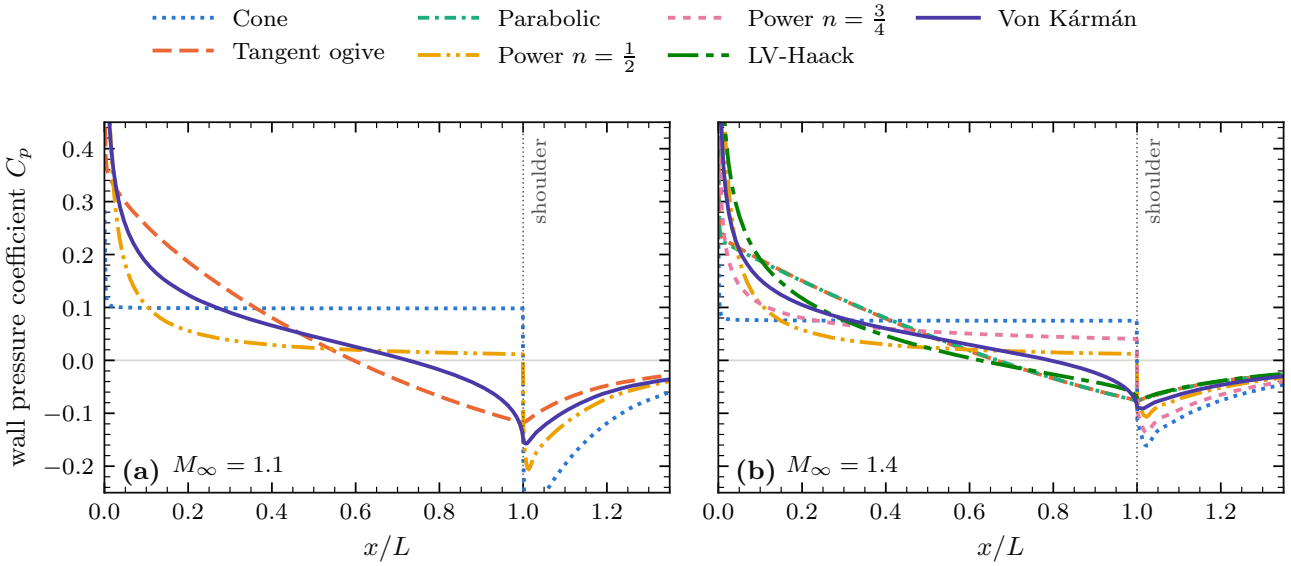


Figure 4: Wall pressure coefficient along the nose ($L/D = 4$) at (a) $M = 1.1$ (only the shapes computed at that Mach number) and (b) $M = 1.4$. The tangent profiles reach $C_p < 0$ over the aft third; the power-$\frac{1}{2}$ nose keeps a small overpressure up to the shoulder after a spike at its blunt tip; the cone keeps the conical value.

Length is as powerful as shape. Going from $L/D = 4$ to 4.5 lowers the Von Kármán wave drag at $M = 1.4$ by 19 % (0.0526 to 0.0425), close to the factor $(4/4.5)^2 = 0.79$ of Equation (2), while at equal length the ogive carries 21 % more than the Von Kármán nose. The design question is therefore how to combine the best profile with the longest nose the vehicle can accommodate.

4.2 Flight performance

Weisshorn burns out at 9.1 s near 2.2 km at $M_{\max} = 1.47$ (nominal) or 1.56 (envelope), spends 16 s above $M = 0.8$ and 10 s above Mach 1, and crosses the transonic band twice (Figure 5a). The drag impulse $\int D\,\mathrm{d}t$, the momentum lost to drag, is spent mostly between $M = 0.8$ and 1.5 (Figure 5b), where the dynamic pressure is highest and where the nose wave drag acts. The apogee sensitivities around the baseline are **–100 m per** $\Delta C_D = 0.01$ and **–143 m/kg**. A wave-drag difference only acts above $M = 0.8$ and is worth somewhat less: the ogive carries 0.009 more $C_{D,\mathrm{w}}$ than Von Kármán at $M = 1.4$ and $L/D = 4.5$, and loses 79 m rather than 90 m. Skin friction dominates the vehicle drag at all speeds (Weisshorn is a fineness-30 airframe), base drag is second, and the nose wave drag is 10–12 % of the total above $M = 0.8$. This is why the nose shape moves the apogee by tens of metres rather than kilometres, and why the length (wetted area) and shape effects are of the same order.

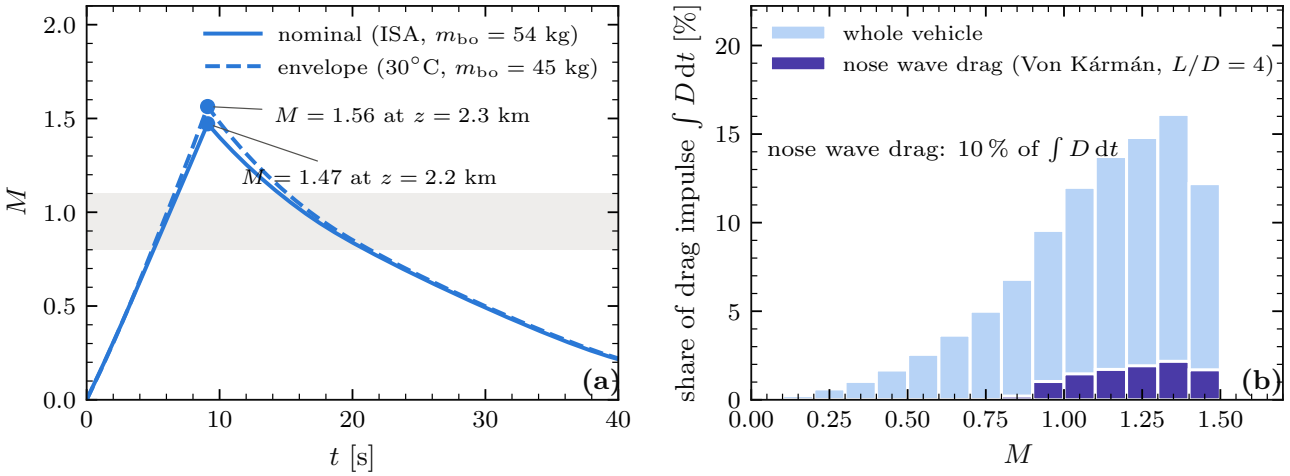


Figure 5: (a) Mach number along the nominal and hot-day envelope ascents (grey band: transonic range). (b) Share of the drag impulse per Mach bin for the baseline nose; the dark part is the nose wave drag.

4.3 Trade study

Figure 6 gives, for every profile and fineness, the apogee change relative to the baseline, the length over which a 3U section fits, and the change in static margin; Table 3 lists the candidates at and near the LVS length. Each point of Figure 6a is a full trajectory with the wave drag, length, wetted area and shell mass of that nose. The curves rise with fineness because the wave drag falls as $1/(L/D)^2$, and flatten because the friction and mass of the longer nose grow linearly with L.

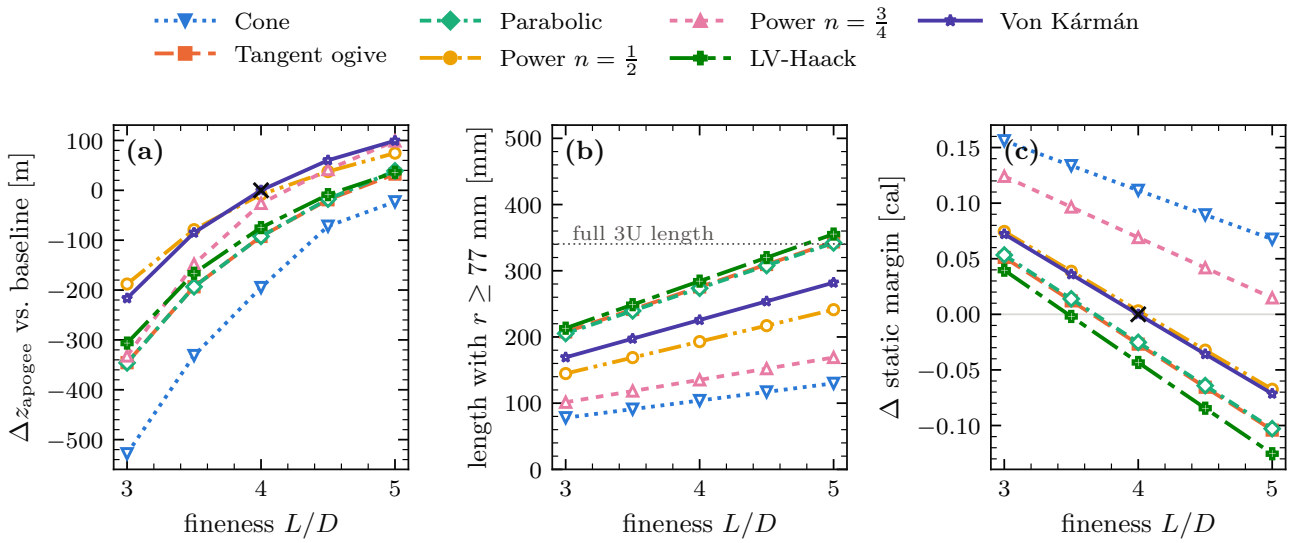


Figure 6: (a) Apogee change relative to the baseline (Von Kármán, $L/D = 4$, cross), including the extra length, wetted area and mass of longer noses. (b) Length of the curved nose with $r \geq 77$ mm (3U fit). (c) Change in subsonic static margin.

Table 3: Candidates at $L/D = 4$ (LVS length, 720 mm) and $L/D = 4.5$ (810 mm). Δz: apogee change vs. the baseline (Von Kármán, $L/D = 4$); mass: shell + Al tip; 3U in nose: curved length with $r \geq 77$ mm; bay: curved length plus the cylinder the 3U still needs.

L/D	Profile	$C_{D,\text{w}}$ at $M =$ 1.1	1.2	1.4	Δz [m]	mass [g]	3U in nose [mm]	bay [mm]	ΔSM [cal]
4	Cone	0.0982	0.0862	0.0750	-195	261	104	966	+0.11
4	Tangent ogive	0.0651	0.0657	0.0640	-93	328	275	795	-0.03
4	Parabolic	0.0656	0.0656	0.0639	-92	328	274	797	-0.03
4	Power $n = \frac{1}{2}$	0.0523	0.0541	0.0550	-8	328	193	878	+0.00
4	Power $n = \frac{3}{4}$	0.0628	0.0628	0.0564	-26	289	135	935	+0.07
4	LV-Haack	0.0610	0.0610	0.0613	-75	341	284	786	-0.04
4	**Von Kármán**	0.0541	0.0541	0.0526	+0	321	226	845	+0.00
4.5	Cone	0.0708	0.0708	0.0619	-72	286	117	1043	+0.09
4.5	Tangent ogive	0.0530	0.0530	0.0516	-19	361	309	870	-0.07
4.5	Parabolic	0.0529	0.0529	0.0515	-18	361	308	870	-0.06
4.5	Power $n = \frac{1}{2}$	0.0450	0.0450	0.0457	+38	361	217	943	-0.03
4.5	Power $n = \frac{3}{4}$	0.0513	0.0513	0.0463	+42	318	152	1008	+0.04
4.5	LV-Haack	0.0496	0.0496	0.0497	-9	376	320	870	-0.08
4.5	**Von Kármán**	0.0437	0.0437	0.0425	+60	354	254	907	-0.04

- **Von Kármán, power $\frac{1}{2}$ and power $\frac{3}{4}$ lead at every length** and swap places within about 30 m: power $\frac{1}{2}$ is ahead at $L/D \leq 3.5$, Von Kármán at $L/D = 4$–4.5, power $\frac{3}{4}$ at $L/D = 5$. At $L/D = 4$, Von Kármán beats the tangent ogive (Bella Lui II / Nordend) by 93 m and the Firehorn power-$\frac{1}{2}$ by 8 m.
- **Length pays, with diminishing returns**: 60 m per extra 90 mm ($\Delta L/D = 0.5$) around $L/D = 4$–4.5.
- **Fuller noses house more payload**: at $L/D = 4$ the ogive and LV-Haack fit 275 mm to 284 mm of the 3U inside the curved part, against 226 mm for Von Kármán. Measured as nose-bay length

(curved part plus the cylinder still needed for the 3U), Von Kármán still gives the highest apogee per millimetre of bay.

- **Stability is not a discriminator**: all candidates change the static margin by less than 0.1 cal (at most 0.16 cal over the whole design space, against the 2–4 cal target).

4.4 Robustness of the ranking

The weakest parts of the chain are the transonic model, the drag of the rest of the vehicle and the mass model. Figure 7 recomputes the apogee of every shape at $L/D = 4.5$ with the onset of the transonic drag rise moved to $M = 0.75$ and 0.85, the transonic wave drag scaled by $\pm 20\,\%$, the rest of the vehicle drag increased by 20 %, and the hot-day envelope with the LVO masses, relative to Von Kármán in the same scenario:

- **Von Kármán flies highest in every scenario.** The power noses stay within 11 m to 31 m of it, the ogive, parabolic and LV-Haack noses are 57 m to 98 m lower, and the cone 98 m to 156 m lower; no design crosses zero.
- The spread between scenarios is smaller than the gap between the groups, so the selection does not hinge on the transonic assumption. The hot-day envelope amplifies the differences; a draggier vehicle shrinks them, because the nose is then a smaller share of the total.

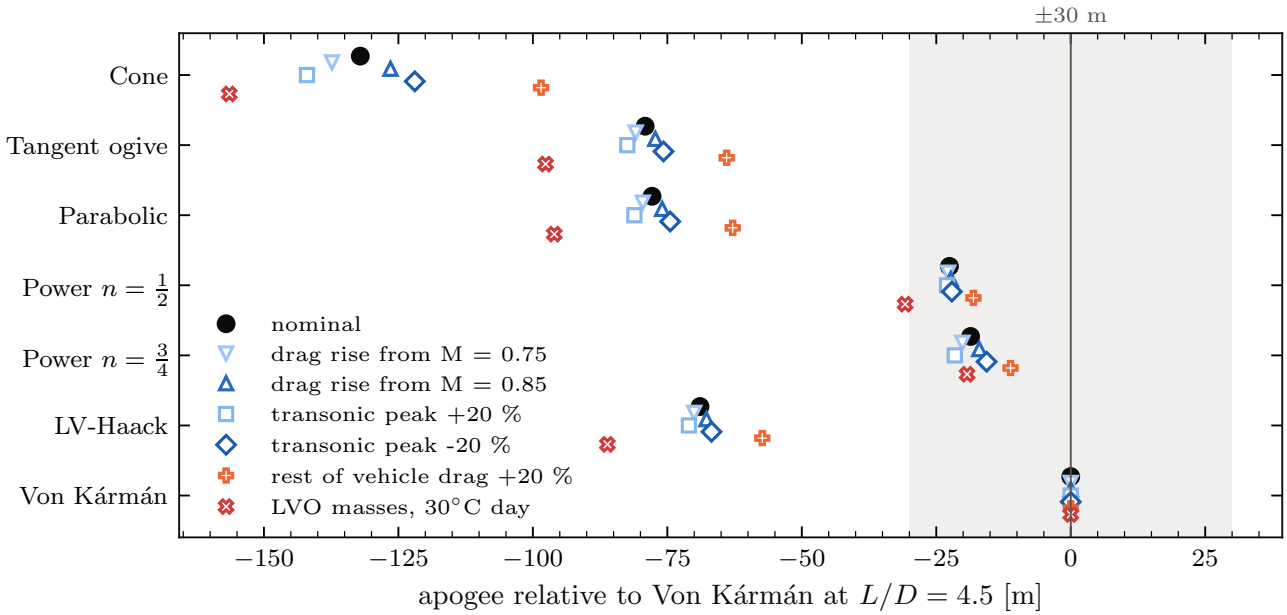


Figure 7: Apogee of each design at $L/D = 4.5$ relative to Von Kármán, under the nominal model and six perturbed ones. The grey band is ±30 m, about ±0.2 kg of dry mass, the mass uncertainty at this stage: designs inside it are equivalent in practice.

4.5 Tip bluntness

A sharp tip cannot be laid up and burns or chips first, so the tip is a separate machined part. On the Von Kármán nose at $M = 1.4$, a 4.5 mm spherical tip ($r_n/R = 0.05$) costs $+0.6\,\%$ in wave drag (–2 m of apogee) and a 9 mm tip ($r_n/R = 0.1$) costs $+3.7\,\%$ (–10 m) (Figure 8a). The cost grows quickly because the stagnation pressure at the centre of the sphere ($C_p \approx 1.5$ at

$M = 1.4$, against about 0.04 averaged over a sharp nose) acts on a growing area. The benefit is a stagnation heat-transfer coefficient falling only as $r_n^{-1/2}$ (Figure 8b): about 2 kW/(m^2 K) for 4.5 mm, and only $\sqrt{2}$ lower at twice the radius. A 4.5 mm tip therefore keeps the drag cost below 1 %, and the heating is handled by the tip material (Section 4.6).

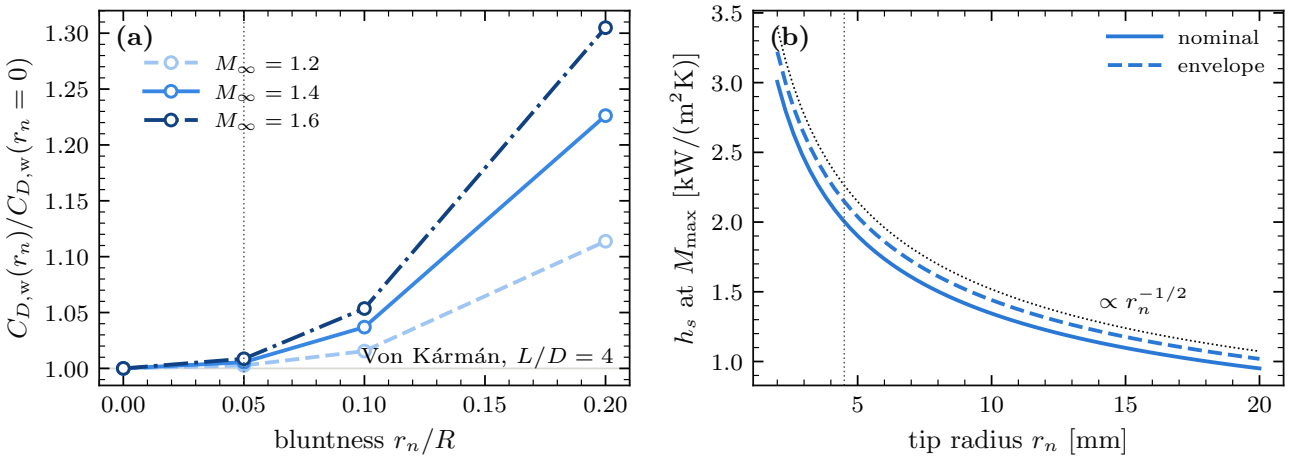


Figure 8: (a) Wave drag of the blunted Von Kármán nose relative to the sharp one. (b) Stagnation-point heat-transfer coefficient at $M_{\max}$ versus tip radius (dotted line: the selected 4.5 mm).

4.6 Heating

The peak stagnation temperature is 119 °C (nominal) and 156 °C (envelope), reached at burn-out (Figure 9).

- **Tip.** A 3D-printed PET-CF tip (Firehorn) conducts heat 600 times less well than aluminium (0.3 against 180 W/(m K)), so its surface follows T_0 and reaches 104 °C (nominal) and 138 °C (envelope), well above the 70 °C to 80 °C heat-deflection range of PET/PETG-CF. A 60 g Al 6082-T6 tip spreads the heat through its whole mass and stays below 51 °C (nominal) and 73 °C (envelope).
- **Shell.** The 1.2 mm CFRP shell follows the recovery temperature within a few seconds. Its inner face lags behind and peaks at 77 °C (nominal) and 104 °C (envelope) about 3 s after burn-out. This is below the cure temperature of the prepreg, but the envelope case comes close to the DP490 coupler bond (shear strength 2.8 MPa at 120 °C). The Firehorn requirement (390 K for 5 s) does not cover the envelope; Section 9 proposes 430 K for 10 s.

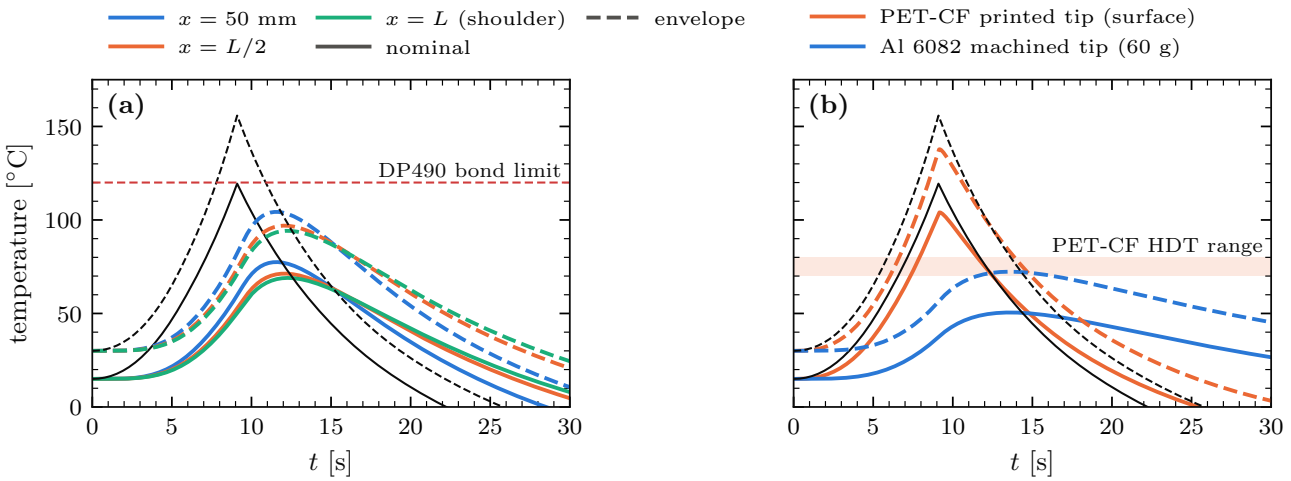


Figure 9: (a) Inner-face temperature of the CFRP shell at three stations (adiabatic inner face, conservative). (b) Tip temperature: printed PET-CF surface against a machined Al 6082 insert. Solid: nominal ascent; dashed: hot-day envelope; black: stagnation temperature T_0.

Heating is a material question, not a shape question: the stagnation temperature is set by the trajectory, the same for every profile. The design protects the temperature-sensitive parts, the tip polymer and the adhesive bond, with a metal tip and a thermal requirement that covers the hot-day envelope.

5 Design Catalogue

This section presents the seven candidates as designs at the recommended length, $L/D = 4.5$ ($L = 810$ mm, $D = 180$ mm), each with the 60 mm bonded shoulder of the Firehorn design. Figure 10 shows them to scale with the 3U payload placed as far forward as each profile allows, Figure 11 the computed flow around them at $M = 1.4$, and Table 4 collects the numbers (wave drag from the direct Euler runs at $L/D = 4.5$).

In Figure 11 the shock leans back at roughly the Mach angle ($45.6°$, Equation (1)). On the cone and the power noses the expansion is concentrated in a fan at the shoulder kink; on the tangent profiles it is spread over the aft part of the nose itself. The more intense the red region in contact with the nose, and the less blue on it, the higher the wave drag. The designs compare as follows.

- **Conical – rejected.** Smallest volume (6.9 L), wetted area and shell (286 g), and the simplest (developable) mould, but a $6.3°$ shoulder kink. The conical overpressure acts on the whole surface and all the turning happens in one expansion at the shoulder, so it has the highest wave drag at every Mach number (46 % above Von Kármán at $M = 1.4$). Only 117 mm of the 3U fits inside (bay 1043 mm); -132 m against Von Kármán.
- **Tangent ogive – acceptable alternative for volume-driven designs.** The classic subsonic nose of Bella Lui II and Nordend, with a $13°$ tip semi-angle and a tangent shoulder. Its fuller forebody compresses the flow harder near the tip: 21 % more wave drag than Von Kármán at $M = 1.4$. It packages well (309 mm of the 3U inside, bay 870 mm) and its single radius is the easiest smooth profile to make, but it costs 79 m of apogee (about 0.55 kg of dry mass) for 37 mm less bay.

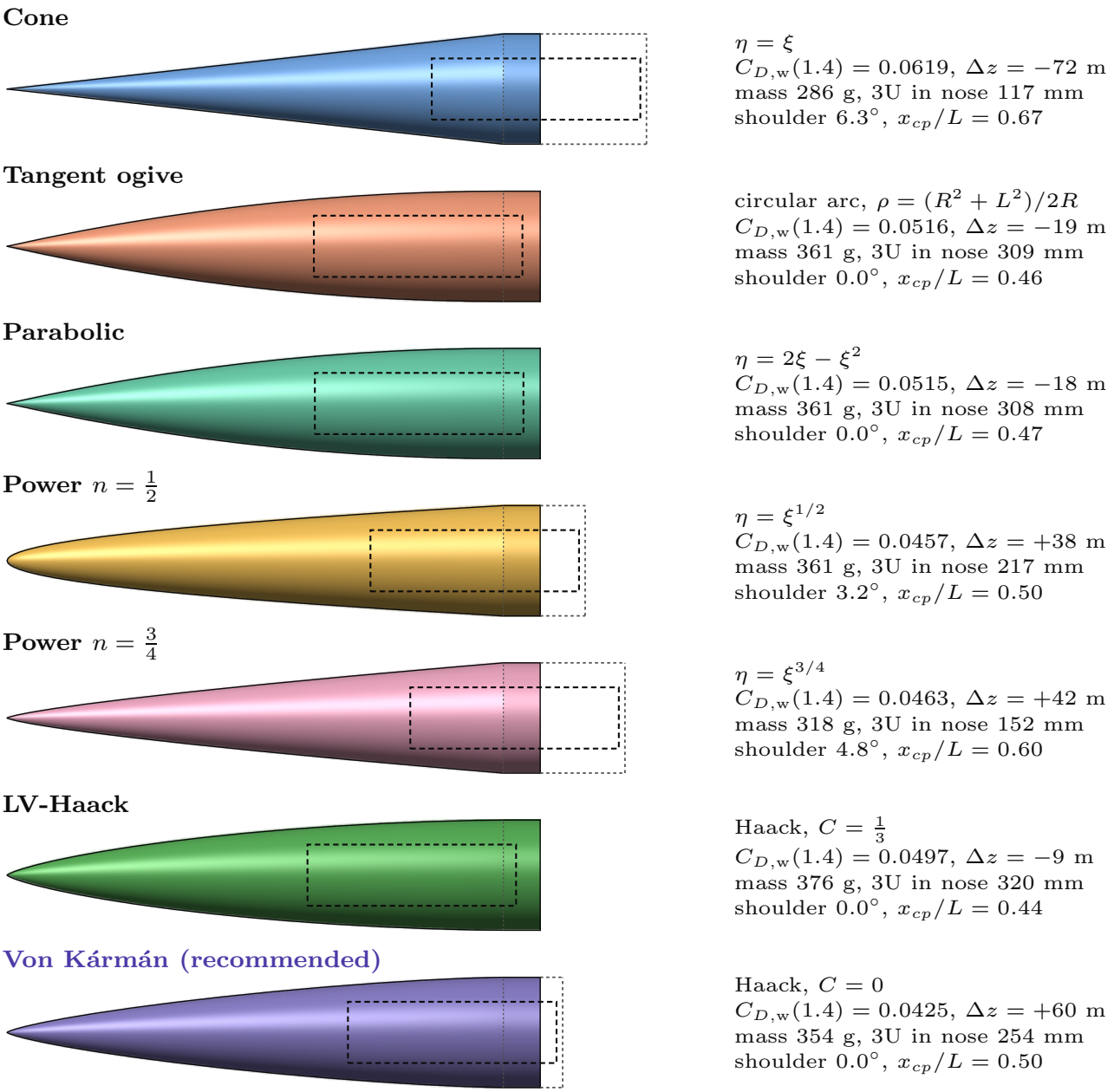


Figure 10: The seven designs at $L/D = 4.5$, rendered to the same scale (diameter 180 mm). Thin dotted line: end of the curved part; dashed box: 3U CubeSat ($100 \times 100 \times 340.5$ mm) pushed forward until its corners reach $r = 77$ mm; dotted extension: payload-bay cylinder still needed behind the 60 mm shoulder. Δz: apogee change relative to the LVS-length baseline (Von Kármán at $L/D = 4$).

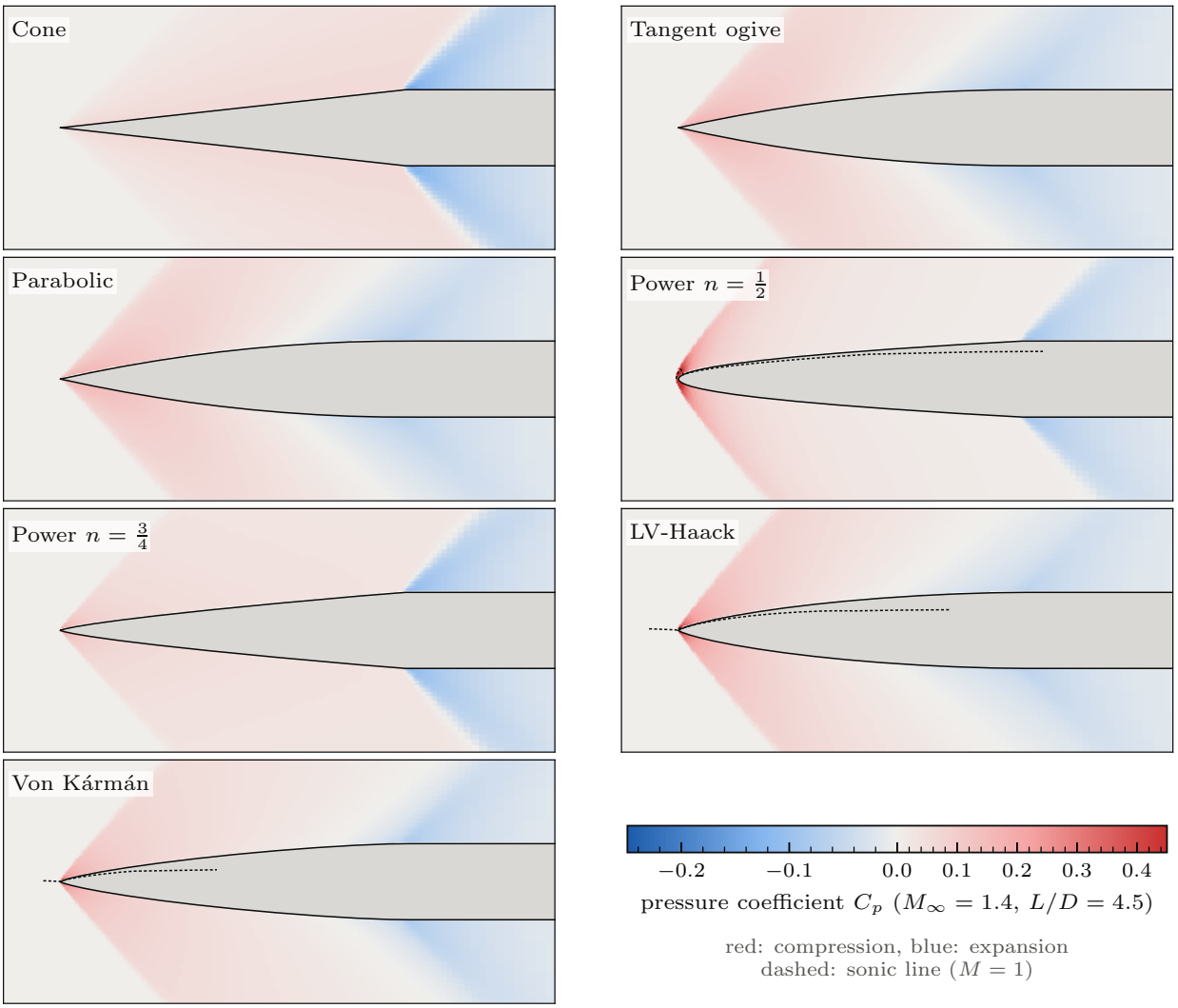


Figure 11: Pressure coefficient around the seven designs at $M_\infty = 1.4$, $L/D = 4.5$ (axisymmetric Euler, mirrored about the axis). The bow shock is the edge of the red region; the blue fans are the expansions that turn the flow back along the cylinder. The dashed line marks the sonic line where a subsonic region exists.

- **Parabolic ($K' = 1$) – same as the ogive.** At equal length it is practically the tangent ogive (wave drag within 0.4 %). Crowell's chart rates it best in the high subsonic range (Table 5), which is irrelevant for a vehicle whose drag losses lie at $M = 0.8$–1.5.
- **Power $n = \frac{1}{2}$ (Wildhorn, Firehorn) – the choice for a short nose.** Same volume as Von Kármán (10.3 L) but further forward, and a $3.2°$ shoulder kink. The blunt apex detaches the bow shock and creates the only closed sonic line of Figure 11; behind this spike the wall pressure stays slightly above ambient. Its wave drag is 3–10 % above Von Kármán at $L/D = 4.5$, but it is the lowest of all at $L/D = 3$ and at $M \leq 1.1$ for $L/D = 4$. 217 mm of the 3U fits inside (bay 943 mm); it flies 22 m below Von Kármán (about 0.15 kg) and within 31 m of it in every robustness scenario.
- **Power $n = \frac{3}{4}$ – not selected.** Between the cone and the $\frac{1}{2}$-power nose: pointed tip, slender body, small volume (8.2 L), $4.8°$ kink. Its wave drag falls fastest with Mach number ($0.0513 \rightarrow 0.0432$ from $M = 1.2$ to 1.6) and its low wetted area saves friction, so it ends only 18 m below Von Kármán. It has the worst packaging of the smooth shapes (152 mm of the 3U inside, bay 1008 mm) and would only make sense on a faster vehicle ($M_{\max} \approx 2$) with the payload outside the nose.
- **LV-Haack ($C = \frac{1}{3}$) – the choice if payload volume were the driver.** The minimum wave drag for a given length *and volume*: the fullest design (11.6 L), tangent shoulder, 13–19 % more wave drag than Von Kármán over $M = 1.2$–1.6. It has the best payload room (320 mm of the 3U inside, bay 870 mm), the heaviest shell (376 g) and the largest static-margin reduction (-0.08 cal); it costs 69 m, marginally better than the ogive for the same bay.
- **Von Kármán ($C = 0$) – recommended.** The minimum wave drag for a given length *and base area*: not constrained in volume, it stays thin near the tip and spreads the area growth along the length. 10.3 L, tangent shoulder, $x_{cp}/L = 0.5$. The compression is weak and confined to the first few per cent of the length, and the wall pressure is below ambient over the aft third. It has the lowest wave drag at every computed Mach number at $L/D = 4.5$ (0.0437, 0.0425 and 0.0414 at $M = 1.2$, 1.4 and 1.6); the 4.5 mm tip adds $+0.5$ to $+1.4\,\%$. 254 mm of the 3U fits inside (bay 907 mm), the tangent shoulder gives a clean bond line, and it flies highest in all seven robustness scenarios.

Table 4: Design catalogue at $L/D = 4.5$ ($L = 810\,\text{mm}$). Wave drag from the direct Euler runs at this fineness; Δz relative to Von Kármán at $L/D = 4$ (LVS length), including length and mass effects.

	Cone	Ogive	Parabolic	Power $\frac{1}{2}$	Power $\frac{3}{4}$	LV-Haack	Von Kármán
Volume [L]	6.87	11.03	10.99	10.31	8.24	11.59	10.31
Wetted area [m^2]	0.230	0.308	0.307	0.307	0.263	0.322	0.300
Tip semi-angle [°]	6	13	13	89	74	80	78
Shoulder angle [°]	6.3	0.0	0.0	3.2	4.8	0.0	0.0
x_{cp}/L (Barrowman)	0.667	0.465	0.467	0.500	0.600	0.438	0.500
$C_{D,\text{w}}$ at $M = 1.2$	0.0708	0.0530	0.0529	0.0450	0.0513	0.0496	0.0437
$C_{D,\text{w}}$ at $M = 1.4$	0.0619	0.0516	0.0515	0.0457	0.0463	0.0497	0.0425
$C_{D,\text{w}}$ at $M = 1.6$	0.0567	0.0504	0.0502	0.0456	0.0432	0.0491	0.0414
Δz at $L/D = 4$ [m]	-195	-93	-92	-8	-26	-75	+0
Δz at $L/D = 4.5$ [m]	-72	-19	-18	+38	+42	-9	+60
Δz at $L/D = 5$ [m]	-23	+32	+39	+75	+99	+36	+99
Shell + tip mass [g]	286	361	361	361	318	376	354
3U inside nose [mm]	117	309	308	217	152	320	254
Nose-bay length [mm]	1043	870	870	943	1008	870	907
Δ static margin [cal]	+0.09	-0.07	-0.06	-0.03	+0.04	-0.08	-0.04

5.1 Literature cross-check

The drag-rating chart compiled by Crowell (reproduced on Wikipedia's "Nose cone design" page, CC BY-SA 4.0, and used in the Firehorn nosecone DJF) condenses published drag comparisons into four qualitative classes and gives an independent, if coarse, check (Table 5; the chart does not state the fineness ratio of its data).

Table 5: Crowell's qualitative drag ratings (1 superior, 2 good, 3 fair, 4 inferior; –: no data), read from the vector chart `Nose_cone_drag_comparison.svg` (G. A. Crowell / C. M. G. Lee). Compared with the ranking computed here at $L/D = 3$ and $L/D = 4.5$.

	Crowell rating at $M =$					Euler rank at $M = 1.4$	
Shape	0.9	1.0	1.2	1.5	1.85	$L/D = 3$	$L/D = 4.5$
Cone	4	4	4	4	4	6	7
Tangent ogive	4	4	4	4	4	5	6
Parabola	1	3	–	–	–	4	5
Power $\frac{1}{2}$	1	2	2	2	3	1	2
Power $\frac{3}{4}$	2	–	–	3	2	–	3
LV-Haack	1	1	–	–	–	3	4
Von Kármán	1	1	1	3	2	2	1

Both sources rate the ogive and the cone worst at transonic and low supersonic speeds and the Von Kármán, Haack and power-law noses as the good ones. Around $M \approx 1.5$ the chart prefers the $\frac{1}{2}$-power nose; the Euler results reproduce that at $L/D = 3$ but not at $L/D \geq 4$, where Von Kármán is lower at every Mach number. The ranking depends on fineness, so the recommendation rests on the computed ranking at the actual design fineness.

6 Selection

6.1 Profile

Table 6 scores the profiles at $L/D = 4.5$ from 1 (worst) to 5 (best) and sums the scores with weights in the ratio 4:2:2:1. Performance (apogee change), integration (3U length inside the curved part) and stability (static-margin change) are scored linearly between the worst and the best design; manufacture is a judgement score (a tangent shoulder and a standard mould score 5, a kink is penalised, the cone gets 3 for its simple mould). Performance carries the largest weight (0.44) because apogee margin is what the nose shape mainly affects; integration and manufacture (0.22 each) decide whether the design can be housed and built; stability (0.11) matters least, since the margin varies by less than 0.1 cal. Heating is left out because, with a metallic tip, it does not depend on the profile. Only properties of the hardware are scored: whether a profile has flown on an earlier ERT vehicle is not a criterion.

Table 6: Trade-off matrix at $L/D = 4.5$ (1 = worst, 5 = best). Performance: apogee at equal length (Table 3). Integration: 3U length inside the curved part. Manufacture: tangent shoulder, mould and tip. Stability: change in static margin.

Criterion (weight)	Cone	Ogive	Parab.	Power $\frac{1}{2}$	Power $\frac{3}{4}$	LV-Haack	Von Kármán
Performance (0.44)	1.0	2.6	2.6	4.3	4.4	2.9	5.0
Integration, 3U fit (0.22)	1.0	4.8	4.8	3.0	1.7	5.0	3.7
Manufacture (0.22)	3.0	5.0	5.0	3.0	2.0	5.0	5.0
Stability (0.11)	5.0	1.4	1.5	2.2	3.9	1.0	2.1
Weighted total	1.89	3.49	3.51	3.49	3.23	3.63	**4.39**

Von Kármán is selected. It has the lowest wave drag at every computed Mach number, flies 18 m higher than power $\frac{3}{4}$ and 22 m higher than power $\frac{1}{2}$, and stays first in all seven robustness scenarios; ogive, parabolic and LV-Haack lose 69 m to 79 m, and the cone 132 m. Against the two power noses it also wins on integration: a tangent shoulder (no kink at the coupler bond, against $3.2°$ and $4.8°$) and more of the 3U inside (254 mm, against 217 mm and 152 mm). Its advantage holds at $M \geq 1.2$, so it does not hinge on the transonic model.

The outcome does not depend on the exact weights. With equal weights Von Kármán still leads LV-Haack (3.95 against 3.48); with performance and integration swapped it still scores highest, level with LV-Haack (4.10 against 4.09). It loses first place, to LV-Haack, only if payload room carries more than about half of the total weight, i.e. if the payload grows beyond a single 3U unit (Table 8).

6.2 Length

Every 90 mm of nose ($\Delta L/D = 0.5$) around $L/D = 4$–4.5 is worth 60 m of apogee, about 0.4 kg of mass margin, for about 60 mm of nose-bay length (partly recovered because more of the 3U fits inside) and 30 g of shell. The returns diminish: the wave drag falls as $1/(L/D)^2$ while length, wetted area and mass grow linearly, and a further step to $L/D = 5$ would add only about 39 m. With the LV capped at 5.4 m (2027_C_SE_REQ_04) and dry-mass growth expected (2027_C_SE_REQ_06 caps it at 50 kg), the design point is $L/D = 4.5$**,** $L = \mathbf{810\ mm}$, which gives 60 m of margin for 62 mm of bay length. If systems engineering cannot release those 62 mm, the same profile at $L/D = 4$ (720 mm, the LVS length) is the fallback; nothing else changes.

6.3 Tip

A spherical tip of radius $r_n = 4.5\,\text{mm}$ ($r_n/R = 0.05$), machined in Al 6082-T6 as a replaceable insert. It removes the sharp apex that cannot be laid up and chips on landing, costs $+0.6\,\%$ in wave drag (–2 m of apogee), stays below 73 °C in the envelope case (against 138 °C for a printed PET-CF tip), and adds about 60 g at the very front, which helps the static margin. A 9 mm tip would lower the stagnation heating by only $\sqrt{2}$ for –10 m, and the aluminium insert does not need it.

7 Recommendations

7.1 Numbered recommendations

Table 7: Recommendations, owners (ERT teams, see Nomenclature) and timing.

#	Recommendation	Owner	When
R1	Adopt the Von Kármán profile (Haack $C = 0$). Use `nose_profile_final.csv` or the STEP/STL solids (Section 8) as the master geometry.	FD $\rightarrow$ ST	PDR
R2	Book a 907 mm nose bay (810 mm curved + 97 mm cylinder for the 3U, of which 60 mm is the coupler shoulder) in the 5.4 m LV budget. If it does not fit, keep the same profile at $L = 720$ mm (-60 m, bay 845 mm).	SE	PDR
R3	Use a machined Al 6082-T6 tip insert ($r_n = 4.5$ mm, $\sim$80 mm long, $\sim$60 g), threaded onto a ring bonded inside the shell. Do not fly a printed PET-CF tip: it would exceed its heat-deflection range by 25 °C to 70 °C.	ST	CDR
R4	Lay up the shell in CFRP prepreg $[0/45/0]_{SO}$, 1.2 mm, on a male PET-CF mould printed in 2–3 segments, with the joints in the aft half (low curvature). Use GFRP instead if the nose has to be the RF window.	ST (AV, PL)	CDR
R5	Raise the thermal requirement to 430 K for 10 s and qualify the shoulder bond: the DP490 joint at the shoulder sees about 95 °C in the envelope case. Consider a higher-T_g adhesive if the test margin is small.	SE, ST	CDR
R6	Verify with Fluent (RANS, k-ω SST) at $M = 0.95$, 1.2 and 1.47 for the selected nose and the power-$\frac{1}{2}$ alternative, plus one 5° incidence case for $C_{N\alpha}$ and CP, to replace the transonic model of this report.	FD-CFD	before CDR
R7	Put the final geometry and $C_{D,\text{w}}(M)$ into the RocketPy model. Re-check apogee, rail-exit speed and the 2–4 cal static margin once the fins are designed. The nose contributes $C_{N\alpha} = 2$ at 405 mm from the tip.	FD-Sim	PDR
R8	Size the payload structure around a 3U envelope whose front face is 254 mm ahead of the shoulder, inside the $r \leq 77$ mm contour (Figure 12).	PL, ST	CDR
R9	Create the missing Weisshorn nosecone requirements: geometry, bay length, mass ($\leq$ 500 g for the nose structure, Table 10), thermal and tip (Section 9).	SE	PDR

7.2 When to choose another design

The recommendation assumes the current Weisshorn priorities: apogee margin, a 3U payload in the nose bay, and a length cap. If they change, Table 8 gives the design to use. Do not use a cone or a tangent ogive on a supersonic Weisshorn, nor a printed thermoplastic tip, and judge a longer nose by the nose-bay length it needs, which is what the 5.4 m cap constrains.

Table 8: Choice of nose cone if the priorities change.

If . . .	choose	cost relative to the recommendation
the LV cannot grow beyond the LVS length	Von Kármán, 720 mm	-60 m apogee
the payload grows (more volume in the nose)	LV-Haack, 810 mm	-69 m, bay -37 mm, $+22$ g
the nose must be as short as possible ($L/D \leq 3.5$)	Power $\frac{1}{2}$	lowest drag at short length
the vehicle becomes faster ($M_{\max} \approx 2$) and the payload leaves the nose	Power $\frac{3}{4}$ or Von Kármán at $L/D \geq 5$	re-run the study
a subsonic test vehicle (e.g. Aletsch)	Tangent ogive or Von Kármán	shape barely matters below $M = 0.8$

8 Design Definition

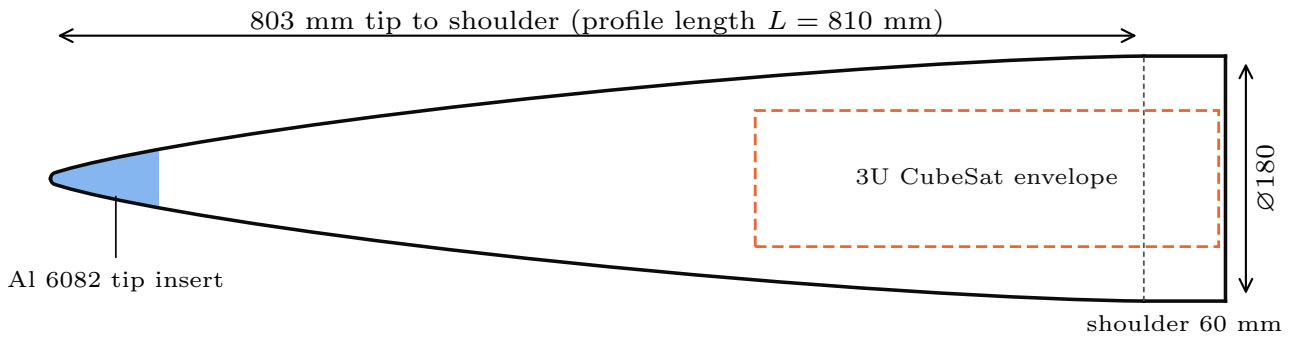


Figure 12: Selected nose cone to scale: Von Kármán profile, $L/D = 4.5$, 4.5 mm spherical Al 6082 tip, 60 mm bonded shoulder; dashed: 3U CubeSat envelope.

Table 9: Nose cone definition (outer mould line and performance).

Profile	Von Kármán (Haack series, $C = 0$): $y = \frac{R}{\sqrt{\pi}}\sqrt{\theta - \frac{1}{2}\sin 2\theta}$, $\theta = \arccos\left(1 - \frac{2x}{L}\right)$
Base diameter $D = 2R$	180 mm
Profile length L (to the theoretical apex)	810 mm ($L/D = 4.5$)
Tip	sphere $r_n = 4.5$ mm, tangent at $x = 9.8$ mm; apex truncated by 6.7 mm
Physical curved length (tip to shoulder)	803.3 mm
Shoulder	60 mm cylinder, Al male coupler bonded inside (Firehorn design)
Angle to the body at the shoulder	$0°$ (tangent)
Internal volume / wetted area	10.31 L / 0.300 m^2
Shell	CFRP prepreg $[0/45/0]_{SO}$, 1.2 mm, on a PET-CF male mould (Firehorn process); GFRP if the nose must be an RF window
Tip insert	Al 6082-T6, machined, threaded onto a bonded Al ring (Matterhorn/Firehorn concept), ~ 60 g, replaceable
Estimated mass (shell + tip, without coupler)	354 g
Nose CP (Barrowman, from the apex)	405 mm, $C_{N\alpha} = 2$
3U inside the curved part	254 mm; the remaining 97 mm of the 3U sits behind the shoulder
Nose wave drag $C_{D,\mathrm{w}}$ at $M = 1.2/1.4/1.6$	0.0437 / 0.0425 / 0.0414 (Euler at $L/D = 4.5$, sharp apex; $+0.5$ to $+1.4$ % with the tip)
Apogee (nominal, model)	9060 m, i.e. 60 m vs. the LVS-length baseline; $M_{\max} = 1.47$
Static margin change vs. baseline	-0.04 cal
Peak temperatures (envelope)	tip 73 °C; shell inner face 104 °C

The 4.5 mm sphere replaces the theoretical apex and removes 6.7 mm of length, hence the 803.3 mm physical curved length. A radius tolerance of ± 0.5 mm moves $C_{D,\mathrm{w}}$ by much less than 1 %; a step at the tip/shell joint matters more, because a forward-facing step acts as a small blunt face with a local shock and heating peak. Keep it below 0.2 mm and forward-facing steps at zero.

8.1 Manufacturing plan

1. **Mould.** Male mould printed in PET-CF from the CAD (Firehorn process). The 870 mm part (curved length + shoulder) is longer than one print bed and is printed in three segments, joined at $x \approx 330$ and 600 mm where the radius changes slowly, aligned on a central rod, sanded and sealed with release agent.
2. **Layup.** Prepreg $[0/45/0]_{SO}$ (1.2 mm), the Firehorn laminate. Plies are cut from flat patterns and stop at $x \approx 60$ mm (radius about 17 mm), which avoids the ply pile-up at the apex reported on Firehorn. Two extra local plies cover the last 100 mm of the curved part and the shoulder,

where the coupler is bonded.

3. **Tip.** Machined Al 6082-T6 insert covering the first 80 mm, threaded onto an Al ring bonded inside the shell at $x = 60$ mm, flush with the shell within 0.2 mm.
4. **Coupler.** Firehorn male glued coupler, scaled to 180 mm, bonded on the 60 mm shoulder (2027_C_ST_COUPLERS).
5. **Finish.** Sand and paint. Check the profile with a printed template (or a CMM) at five stations against `nose_profile_final.csv`, tolerance ± 0.5 mm.

8.2 Mass budget

Table 10: Mass budget of the nose cone structure (without payload, payload bay and coupler).

Item	Mass [g]	Basis
CFRP shell, 0.300 m^2 at 0.98 kg/m^2	294	Firehorn as-built areal density
Local reinforcement plies (last 160 mm)	35	2 plies over 0.09 m^2
Al 6082-T6 tip insert	60	machined, hollowed
Bonded Al tip ring + fastener	20	estimate
Adhesive (tip ring, coupler)	25	estimate
Paint and filler	15	0.05 kg/m^2
Total	**449**	
Proposed requirement	500	11 % margin

8.3 CAD deliverables

The tool of Appendix A writes the profile coordinates (`nose_profile_final.csv`, 200 points, x from the tip), watertight binary STL solids (mm, x from the tip, closed by a flat base) of the seven candidates at $L/D = 4.5$ and of the final design (`nose_vonkarman_LD4.5_rn4.5mm_FINAL.stl`, volume within 0.05 % of the analytical one), and a STEP solid of the final design (same name, `.step`) for CAD and the Ansys runs of R6.

9 Open Points and Next Steps

1. **Fluent verification (CFD team).** Run the selected geometry in Fluent (density-based, k-ω SST, ideal gas, Sutherland viscosity) at $M = 0.95$, 1.2 and 1.47, plus the 5° incidence case (R6). This replaces the transonic assumption of Section 3.2 and checks the Euler wave drag with viscous effects; the Euler wall-pressure files are the reference to compare against. The setup is detailed in `CFD/Fluent_Workflow_Nose_Cone.pdf` of the repository (Appendix A).
2. **RocketPy update (simulation team).** Replace the LVS nose in the RocketPy model by the final geometry and its $C_D(M)$, then re-check apogee, the 30 m/s rail exit speed and the static margin (2–4 cal, 2027_C_SE_FD_REQ_07) once the fins are defined.
3. **Requirements to create** (none exists for the Weisshorn nosecone). Proposed:
   - profile: Von Kármán, $L = 810$ mm ($L/D = 4.5$, 803.3 mm physical with the tip), coordinates from `nose_profile_final.csv`, radius tolerance ± 0.5 mm;

- nosecone bay length (curved part + the cylinder that completes the 3U housing) 907 mm, to be booked in the 5.4 m LV budget (2027_C_SE_REQ_04);
- thermal: withstand 430 K on the outer surface for 10 s (envelope stagnation temperature, an upper bound for the whole surface), replacing Firehorn's 390 K/5 s;
- tip: metallic (Al 6082-T6), spherical radius 4.5 mm, replaceable.

4. **Material and RF.** If the payload or avionics need GNSS/telemetry through the nose, the shell must be GFRP (2027_C_SE_AV_INTERFACE_ST_REQ_07 asks for a $360°$ RF window $\geq 2D$ long; the 810 mm nose can be that window). GFRP has the same epoxy temperature limit but about 40 % more mass for the same stiffness; the shape result does not change.
5. **Thermal test.** Update 2025_C_ST_NOSECONE_TEMPERATURE_TSP to the new level and add a shoulder/coupler bond specimen: the DP490 bond is the weakest point in the envelope case (Section 4.6).

A Scripts and data

The study is implemented in the Python package `nosecone`, in the repository `github.com/Landonowins/weisshorn-nose-cone` (private; ask the author for access). The command

```
python -m nosecone run configs/weisshorn.yaml
```

reproduces every number, table, figure and CAD file of this report from the stored Euler database: `data/euler_results.csv` (every converged Euler point) and `data/cfd/` (wall-pressure distributions and flow fields of every run). `tests/verify_reference.py` checks the reproduction against the original results, `legacy/scripts/` keeps the original one-off scripts, and the user guide (`docs/user_guide/`) explains how to apply the tool to another vehicle. Dependencies: Python ≥ 3.11, NumPy, SciPy, pandas, Matplotlib, PyYAML, and gmsh for the STEP export.